

Thm. Let $f: [a, b]$ be a monotone increasing function.

Then, for any $x \in (a, b)$, $f(x^-)$ and $f(x^+)$ exist with

$$\sup_{a \leq t < x} f(t) = f(x^-) \leq f(x) \leq f(x^+) \leq \inf_{x < t \leq b} f(t)$$

Moreover, if $a \leq x < y \leq b$, then $f(x^+) \leq f(y^-)$.

Proof. Since f is mono. increasing, for any $t \in (a, x)$, $f(t) \leq f(x)$.

Therefore, $\sup_{a \leq t < x} f(t) = \sup \{f(t) \mid a \leq t < x\} \leq f(x)$.

Given $\epsilon > 0$, there exists $\delta > 0$ such that

$$\sup_{a \leq t < x} f(t) - \epsilon < f(x - \delta) \leq \sup_{a \leq t < x} f(t)$$

Now, for any $s \in (x - \delta, x)$, $f(x - \delta) \leq f(s)$, therefore

$$\sup_{a \leq t < x} f(t) - f(s) < \epsilon$$

so, $f(x^-) = \sup_{a \leq t < x} f(t)$]

Meanwhile, the second assertion can be proved by:

$$f(x^+) = \inf_{x < t \leq b} f(t) = \inf_{x < t \leq y} f(t) \quad \text{and} \quad f(y^-) = \sup_{a \leq t < y} f(t) = \sup_{a \leq t < x} f(t)$$

□

Corollary. A discontinuous monotone function is at most countable.

Proof. Let $f: (a, b) \rightarrow \mathbb{R}$ be a monotone increasing function, E be the set of discontinuity points. Since f is monotone, $x \in E$ if and only if $f(x^-) < f(x^+)$. By density of rationals, given x ,

$$f(x^-) < r(x) < f(x^+)$$

for some rational $r(x) \in \mathbb{Q}$. And if $x_1 < x_2$, then

$$r(x_1) < f(x_1^+) \leq f(x_2^-) < r(x_2)$$

implies $r(x_1) < r(x_2)$. Therefore, there is injective map from E into $\mathbb{Q}$.

Thm. If $f: (a,b) \rightarrow \mathbb{R}$ satisfies Intermediate Value Property,
then f cannot have first-kind discontinuity.

Proof. Suppose that f has first-kind discontinuity at $x \in (a,b)$.

Then, $f(x^-)$ and $f(x^+)$ exist, but $f(x^-) = f(x) = f(x^+)$ cannot be happen.

WLOG, suppose $f(x^-) \neq f(x)$. put $\epsilon = |f(x^-) - f(x)| > 0$.

Since $f(x^-)$ exists, there exists $\delta > 0$ s.t. $t \in (x-\delta, x) \Rightarrow |f(x^-) - f(t)| < \frac{\epsilon}{3}$

Now, if $f(x) > f(x^-)$, then

$$f(x - \frac{\delta}{2}) < f(x - \frac{\delta}{2}) + \frac{2}{3}\epsilon < f(x)$$

But there is no $c \in (x - \frac{\delta}{2}, x)$ s.t. $f(c) = f(x - \frac{\delta}{2}) + \frac{2}{3}\epsilon$, because

$|f(c) - f(x - \frac{\delta}{2})| < \frac{\epsilon}{3}$. If $f(x) < f(x^-)$, similarly contradiction.